

Volume 7 – Strategy Engine Design Document

AI Quant Trading Platform

Version: 1.0

Document Type: Strategy Engine Design

Classification: Core Trading Engine

1. Purpose

The Strategy Engine is responsible for converting market intelligence into executable trading decisions.

While the AI Decision Engine evaluates probability and market conditions, the Strategy Engine determines **how** a trade should be executed according to predefined rules.

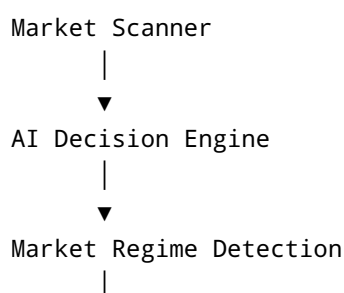
The Strategy Engine must be deterministic, configurable, explainable, testable, and fully compatible with paper trading and backtesting.

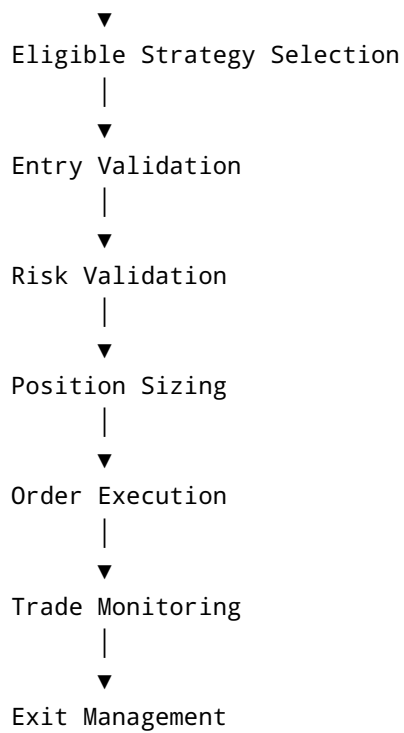
2. Objectives

The Strategy Engine shall:

- Support multiple trading strategies simultaneously
 - Allow users to enable or disable individual strategies
 - Validate all trades through the Risk Engine
 - Prevent conflicting strategy execution
 - Adapt to different market regimes
 - Generate structured trade plans before execution
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3. Strategy Execution Pipeline





4. Strategy Categories

Trend Following

Purpose

Capture medium and long-term directional trends.

Suitable Markets

- Strong Bull Markets
- Strong Bear Markets

Not Suitable

- Sideways Markets

Indicators

- EMA
- ADX
- SuperTrend
- VWAP

Breakout Strategy

Purpose

Trade price breaking important resistance or support.

Confirmation

- Volume Spike
- Candle Close
- Higher Timeframe Trend

Filters

- Fake Breakout Detection
 - Volatility Check
-

Pullback Strategy

Purpose

Join an existing trend after temporary retracement.

Requirements

- Trend Confirmation
 - Pullback Completion
 - Momentum Recovery
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Momentum Strategy

Purpose

Trade assets showing exceptional strength.

Inputs

- Volume
 - RSI
 - MACD
 - ATR
 - Price Acceleration
-

Swing Strategy

Holding Period

Several hours to several days.

Requirements

- Higher Timeframe Trend
 - Strong Support
 - Strong Resistance
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Scalping Strategy

Holding Time

Seconds to Minutes

Requirements

- High Liquidity
- Tight Spread
- Low Slippage

Risk

Highest

Grid Trading

Purpose

Generate profits inside ranging markets.

Requirements

- Sideways Market
- Stable Volatility

Automatically places multiple buy/sell levels.

Dollar Cost Averaging (DCA)

Purpose

Gradually build positions.

Rules

- Maximum Number of Entries
 - Minimum Distance Between Entries
 - Maximum Exposure
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Mean Reversion

Purpose

Trade reversion toward statistical averages.

Indicators

- Bollinger Bands
 - RSI
 - ATR
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News Reaction Strategy

Purpose

React to significant market news.

Sources

- Exchange Announcements
 - Token Listings
 - ETF News
 - Regulatory Updates
-

Smart Money Concepts (SMC)

Concepts

- Liquidity Sweep
 - Order Blocks
 - Fair Value Gap (FVG)
 - Break of Structure (BoS)
 - Change of Character (ChoCH)
 - Premium / Discount Zones
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ICT Concepts

Support

- Optimal Trade Entry
 - Order Blocks
 - Liquidity Pools
 - Market Structure
 - Session Timing
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5. Strategy Selection

The AI Decision Engine determines market conditions.

The Strategy Engine selects only strategies compatible with:

- Market Regime
 - User Risk Profile
 - Timeframe
 - Portfolio Exposure
 - Active Positions
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6. Entry Validation

Every strategy validates:

Trend

Momentum

Volume

Liquidity

Spread

Market Structure

Risk

Higher Timeframe Confirmation

News Filter

Exchange Availability

All conditions must pass before execution.

7. Exit Management

Supported Exit Methods

Fixed Take Profit

Multiple Take Profit Levels

Trailing Stop

Time-Based Exit

Indicator Exit

Manual Exit

Emergency Exit

AI Recommendation Exit

8. Stop Loss Management

Supported Methods

Fixed Percentage

ATR-Based

Swing High/Low

Support & Resistance

Order Block

Volatility Adaptive

Trailing Stop

Break-even Stop

9. Position Sizing

Supported Models

Fixed Size

Fixed Risk %

Kelly Criterion (Experimental)

Volatility-Based

ATR-Based

Portfolio Percentage

Maximum Exposure

10. Trade Filters

Reject trades if:

- Volume is insufficient
 - Spread is excessive
 - Volatility exceeds configured limits
 - Higher timeframe conflicts
 - News blackout period
 - Daily loss limit reached
 - Exchange unavailable
 - Risk Engine rejects trade
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11. Trade Lifecycle

Candidate Generated

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Strategy Selected

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Entry Approved

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Risk Approved



Position Opened



Trade Monitored



Exit Triggered



Performance Stored



Learning Dataset Updated

12. Strategy Performance Tracking

Each strategy records:

- Total Trades
 - Winning Trades
 - Losing Trades
 - Win Rate
 - Average Profit
 - Average Loss
 - Profit Factor
 - Drawdown
 - Sharpe Ratio
 - Average Holding Time
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13. AI Compatibility

The Strategy Engine consumes AI recommendations but remains rule-based.

AI provides:

- Confidence Score
- Risk Estimate

- Market Regime
- Opportunity Ranking

Strategy logic determines whether execution is allowed.

14. Paper Trading Compatibility

Every strategy must support:

- Simulation Mode
- Historical Replay
- Backtesting
- Forward Testing

before live deployment.

15. Configuration

Users may configure:

- Enabled Strategies
 - Timeframes
 - Risk Profile
 - Maximum Trades
 - Trading Sessions
 - Preferred Assets
 - Confidence Threshold
 - Automation Level
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16. Safety Rules

Strategies shall never:

- Override the Risk Engine
 - Ignore exchange restrictions
 - Submit duplicate orders
 - Trade during emergency stop
 - Exceed exposure limits
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17. Future Strategies

Future versions may include:

- Statistical Arbitrage
 - Pair Trading
 - Cross-Exchange Arbitrage
 - Market Making
 - Options Strategies
 - Volatility Strategies
 - AI-Generated Strategies
 - Reinforcement Learning Policies
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18. Design Principles

- Strategies are modular and independently deployable.
- Every strategy must expose explainable entry and exit criteria.
- All strategy decisions are logged and auditable.
- Strategy execution must be deterministic for identical inputs.
- New strategies can be added through a plugin architecture without modifying the core engine.